

Although there are some similarities in construction and mode of operation between depletion-type and enhancement-type MOSFETs, the characteristics of the enhancement-type MOSFET are quite different from anything obtained thus far. The transfer curve is not defined by Shockley's equation, and the drain current is now cut off until the gate-to-source voltage reaches a specific magnitude. In particular, current control in an n -channel device is now effected by a positive gate-to-source voltage rather than the range of negative voltages encountered for n -channel JFETs and n -channel depletion-type MOSFETs.

Basic Construction

The basic construction of the n -channel enhancement-type MOSFET is provided in Fig. 32. A slab of p -type material is formed from a silicon base and is again referred to as the substrate. As with the depletion-type MOSFET, the substrate is sometimes internally connected to the source terminal, whereas in other cases a fourth lead (labeled SS) is made available for external control of its potential level. The source and drain terminals are again connected through metallic contacts to n -doped regions, but note in Fig. 32 the absence of a channel between the two n -doped regions. This is the primary difference between the construction of depletion-type and enhancement-type MOSFETs—the absence of a channel as a constructed component of the device. The SiO_2 layer is still present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of the p -type material. In summary, therefore, the construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET, except for the absence of a channel between the drain and source terminals.

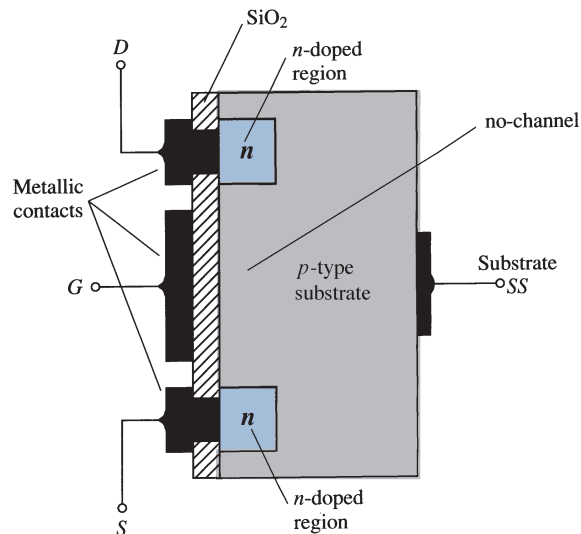


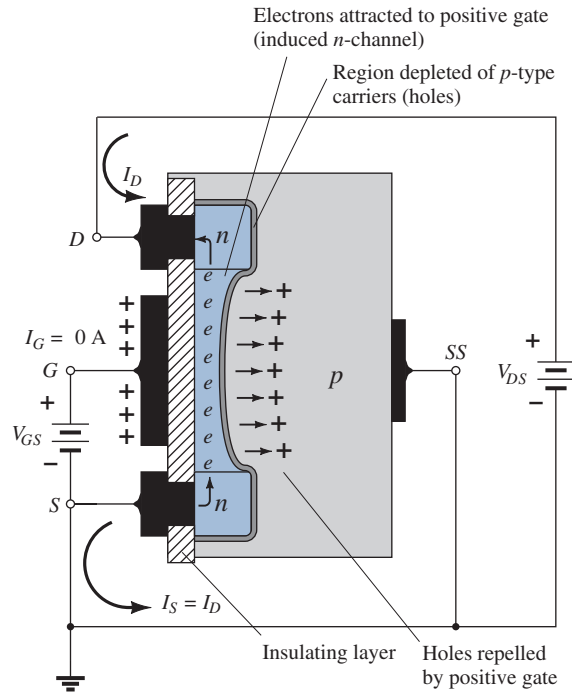
FIG. 32

n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

If V_{GS} is set at 0 V and a voltage applied between the drain and the source of the device of Fig. 32, the absence of an n -channel (with its generous number of free carriers) will result in a current of effectively 0 A—quite different from the depletion-type MOSFET and JFET, where $I_D = I_{DSS}$. It is not sufficient to have a large accumulation of carriers (electrons) at the drain and the source (due to the n -doped regions) if a path fails to exist between the two. With V_{DS} some positive voltage, V_{GS} at 0 V, and terminal SS directly connected to the source, there are in fact two reverse-biased p - n junctions between the n -doped regions and the p -substrate to oppose any significant flow between drain and source.

In Fig. 33, both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and the gate at a positive potential with respect to the source.

**FIG. 33**

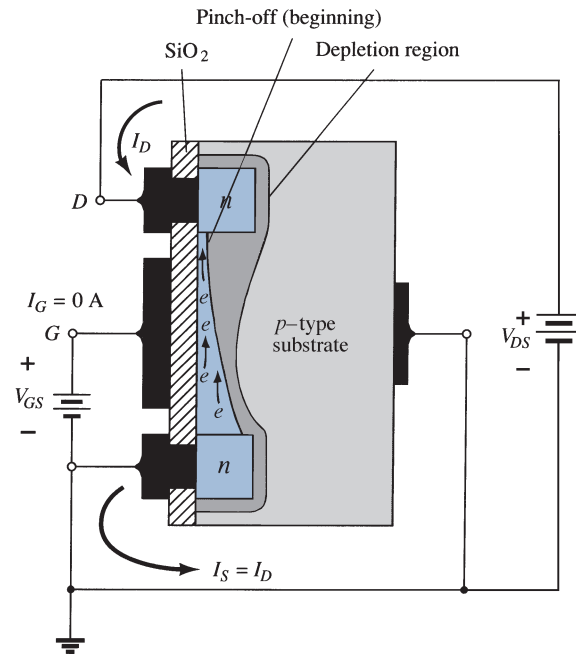
Channel formation in the *n*-channel enhancement-type MOSFET.

The positive potential at the gate will pressure the holes (since like charges repel) in the *p*-substrate along the edge of the SiO₂ layer to leave the area and enter deeper regions of the *p*-substrate, as shown in the figure. The result is a depletion region near the SiO₂ insulating layer void of holes. However, the electrons in the *p*-substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO₂ layer. The SiO₂ layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal. As V_{GS} increases in magnitude, the concentration of electrons near the SiO₂ surface increases until eventually the induced *n*-type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T . On specification sheets it is referred to as $V_{GS(Th)}$, although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with $V_{GS} = 0$ V and “enhanced” by the application of a positive gate-to-source voltage, this type of MOSFET is called an *enhancement-type MOSFET*. Both depletion- and enhancement-type MOSFETs have enhancement-type regions, but the label was applied to the latter since it is its only mode of operation.

As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS} , the drain current will eventually reach a saturation level as occurred for the JFET and depletion-type MOSFET. The leveling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 34. Applying Kirchhoff’s voltage law to the terminal voltages of the MOSFET of Fig. 34, we find that

$$V_{DG} = V_{DS} - V_{GS} \quad (13)$$

If V_{GS} is held fixed at some value such as 8 V and V_{DS} is increased from 2 V to 5 V, the voltage V_{DG} [by Eq. (13)] will increase from -6 V to -3 V and the gate will become less and less positive with respect to the drain. This reduction in gate-to-drain voltage will in turn reduce the attractive forces for free carriers (electrons) in this region of the induced channel, causing a reduction in the effective channel width. Eventually, the channel will be reduced to the point of pinch-off and a saturation condition will be established as described


FIG. 34

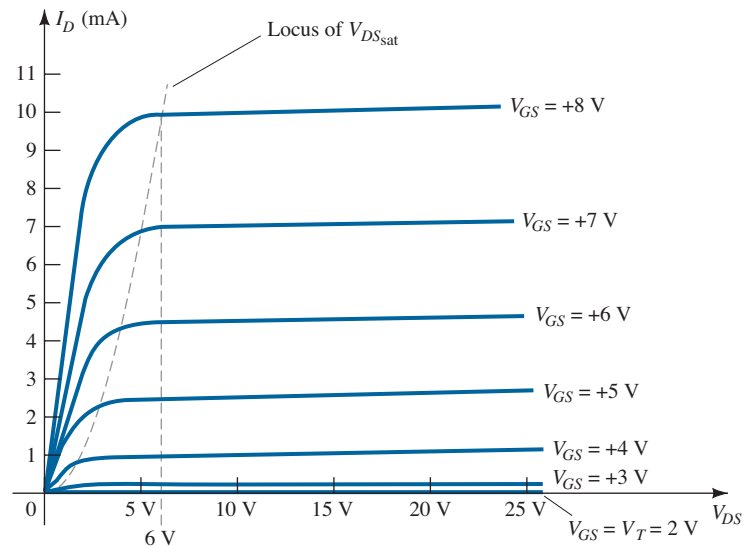
Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS} .

earlier for the JFET and depletion-type MOSFET. In other words, any further increase in V_{DS} at the fixed value of V_{GS} will not affect the saturation level of I_D until breakdown conditions are encountered.

The drain characteristics of Fig. 35 reveal that for the device of Fig. 34 with $V_{GS} = 8$ V, saturation occurs at a level of $V_{DS} = 6$ V. In fact, the saturation level for V_{DS} is related to the level of applied V_{GS} by

$$V_{DS_{sat}} = V_{GS} - V_T \quad (14)$$

Obviously, therefore, for a fixed value of V_T , the higher the level of V_{GS} , the greater is the saturation level for V_{DS} , as shown in Fig. 34 by the locus of saturation levels.


FIG. 35

Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3} \text{ A/V}^2$.

For the characteristics of Fig. 34, the level of V_T is 2 V, as revealed by the fact that the drain current has dropped to 0 mA. In general, therefore:

For values of V_{GS} less than the threshold level, the drain current of an enhancement-type MOSFET is 0 mA.

Figure 35 clearly reveals that as the level of V_{GS} increases from V_T to 8 V, the resulting saturation level for I_D also increases from a level of 0 mA to 10 mA. In addition, it is quite noticeable that the spacing between the levels of V_{GS} increases as the magnitude of V_{GS} increases, resulting in ever-increasing increments in drain current.

For levels of $V_{GS} > V_T$, the drain current is related to the applied gate-to-source voltage by the following nonlinear relationship:

$$I_D = k(V_{GS} - V_T)^2 \quad (15)$$

Again, it is the squared term that results in the nonlinear (curved) relationship between I_D and V_{GS} . The k term is a constant that is a function of the construction of the device. The value of k can be determined from the following equation [derived from Eq. (15)], where $I_{D(on)}$ and $V_{GS(on)}$ are the values for each at a particular point on the characteristics of the device.

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2} \quad (16)$$

Substituting $I_{D(on)} = 10$ mA when $V_{GS(on)} = 8$ V from the characteristics of Fig. 35 yields

$$\begin{aligned} k &= \frac{10 \text{ mA}}{(8 \text{ V} - 2 \text{ V})^2} = \frac{10 \text{ mA}}{(6 \text{ V})^2} = \frac{10 \text{ mA}}{36 \text{ V}^2} \\ &= \mathbf{0.278 \times 10^{-3} \text{ A/V}^2} \end{aligned}$$

and a general equation for I_D for the characteristics of Fig. 35 results in

$$I_D = 0.278 \times 10^{-3}(V_{GS} - 2 \text{ V})^2$$

Substituting $V_{GS} = 4$ V, we find that

$$\begin{aligned} I_D &= 0.278 \times 10^{-3}(4 \text{ V} - 2 \text{ V})^2 = 0.278 \times 10^{-3}(2)^2 \\ &= 0.278 \times 10^{-3}(4) = \mathbf{1.11 \text{ mA}} \end{aligned}$$

as verified by Fig. 35. At $V_{GS} = V_T$, the squared term is 0, and $I_D = 0$ mA.

For the dc analysis of enhancement-type MOSFETs to appear in the chapter “FET Biasing”, the transfer characteristics will again be the characteristics to be employed in the graphical solution. In Fig. 36, the drain and transfer characteristics have been set side by side to describe the

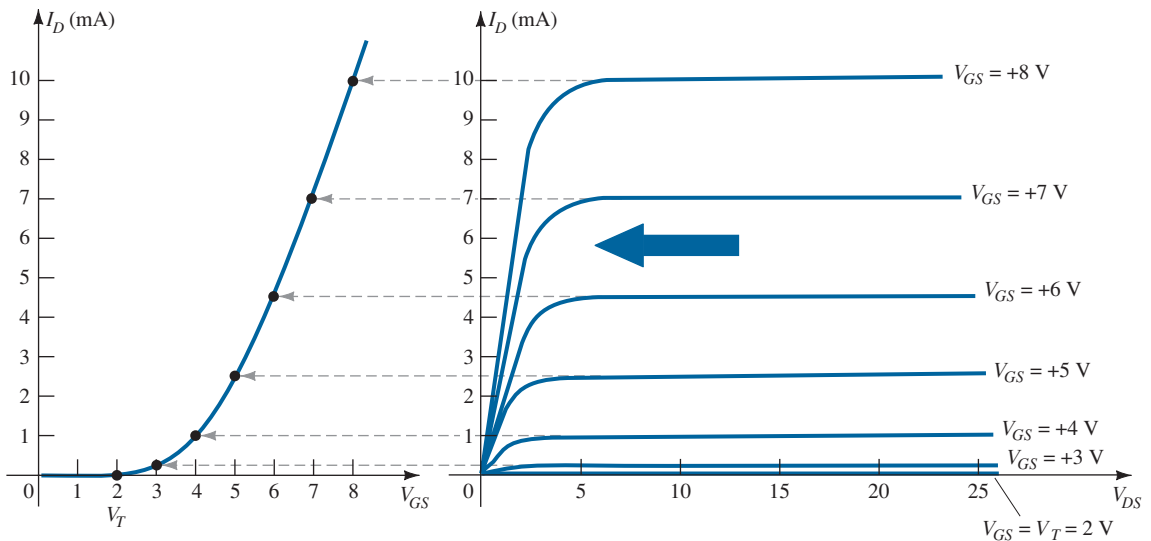


FIG. 36

Sketching the transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics.

transfer process from one to the other. Essentially, it proceeds as introduced earlier for the JFET and depletion-type MOSFETs. In this case, however, it must be remembered that the drain current is 0 mA for $V_{GS} \leq V_T$. As V_{GS} is increased beyond V_T , the drain current I_D will begin to flow at an increasing rate in accordance with Eq. (15). Note that in defining the points on the transfer characteristics from the drain characteristics, only the saturation levels are employed, thereby limiting the region of operation to levels of V_{DS} greater than the saturation levels as defined by Eq. (14).

The transfer curve of Fig. 36 is certainly quite different from those obtained earlier. For an n -channel (induced) device, it is now totally in the positive V_{GS} region and does not rise until $V_{GS} = V_T$. The question now surfaces as to how to plot the transfer characteristics given the levels of k and V_T as included below for a particular MOSFET:

$$I_D = 0.5 \times 10^{-3}(V_{GS} - 4 \text{ V})^2$$

First, a horizontal line is drawn at $I_D = 0$ mA from $V_{GS} = 0$ V to $V_{GS} = 4$ V as shown in Fig. 37a. Next, a level of V_{GS} greater than V_T such as 5 V is chosen and substituted into Eq. (15) to determine the resulting level of I_D as follows:

$$\begin{aligned} I_D &= 0.5 \times 10^{-3}(V_{GS} - 4 \text{ V})^2 \\ &= 0.5 \times 10^{-3}(5 \text{ V} - 4 \text{ V})^2 = 0.5 \times 10^{-3}(1)^2 \\ &= \mathbf{0.5 \text{ mA}} \end{aligned}$$

and a point on the plot is obtained as shown in Fig. 37b. Finally, additional levels of V_{GS} are chosen and the resulting levels of I_D obtained. In particular, at $V_{GS} = 6, 7,$ and 8 V, the level of I_D is 2, 4.5, and 8 mA, respectively, as shown on the resulting plot of Fig. 37c.

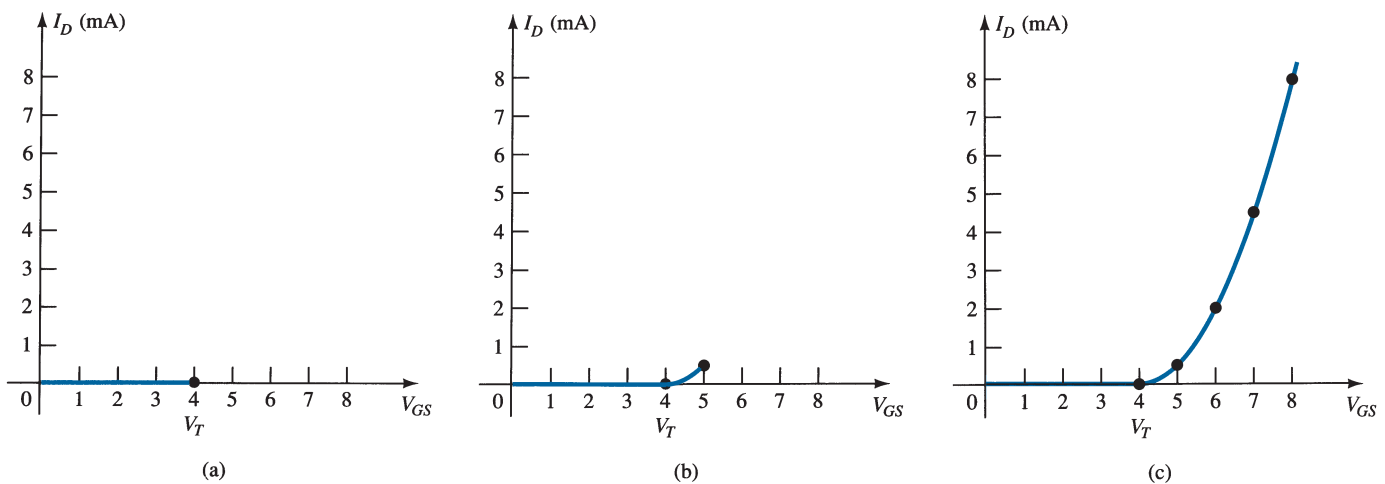


FIG. 37

Plotting the transfer characteristics of an n -channel enhancement-type MOSFET with $k = 0.5 \times 10^{-3} \text{ A/V}^2$ and $V_T = 4 \text{ V}$.

p -Channel Enhancement-Type MOSFETs

The construction of a p -channel enhancement-type MOSFET is exactly the reverse of that appearing in Fig. 32, as shown in Fig. 38a. That is, there is now an n -type substrate and p -doped regions under the drain and source connections. The terminals remain as identified, but all the voltage polarities and the current directions are reversed. The drain characteristics will appear as shown in Fig. 38c, with increasing levels of current resulting from increasingly negative values of V_{GS} . The transfer characteristics of Fig. 38b will be the mirror image (about the I_D axis) of the transfer curve of Fig. 36, with I_D increasing with increasingly negative values of V_{GS} beyond V_T , as shown in Fig. 38c. Equations (13) through (16) are equally applicable to p -channel devices.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for the n - and p -channel enhancement-type MOSFETs are provided as Fig. 39. Again note how the symbols try to reflect the actual construction of